

AMAN KUMAR SINGH

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Summary

Full-stack developer specializing in Java, Node.js, React.js, and MongoDB, with experience building secure and scalable web applications. Proficient in REST API development, authentication systems, and backend engineering, with expanding knowledge of Spring Boot. Strong interest in clean architecture, efficient code, and open-source collaboration.

Education

Master of Computer Application

Sharda University, Uttar Pradesh

PG: 2026

CGPA: 9.27 / 10

Bachelor of Computer Application

JIS College of Engineering, West Bengal

UG: 2024

CGPA: 8.14 / 10

Projects

Air Quality Analysis & Forecasting Dashboard | Python, Flask, LSTM

Feb 2026

- Developed a web-based air quality monitoring system integrating real-time environmental data using APIs to track AQI and pollutant levels (PM2.5, PM10, NO₂, CO, O₃).
- Performed data cleaning and exploratory data analysis (EDA) using **Pandas** and **NumPy** to clean, transform, and analyze air quality datasets.
- Built a **LSTM-based deep learning model** to forecast PM2.5 levels for the next 24 hours, enabling predictive analysis of pollution trends.

EchoSafe Journal | Java, Spring Boot - Backend

Mar 2025

- Architected and implemented a **Spring Boot backend with MongoDB**, applying **RESTful** microservices principles to ensure scalability and high availability.
- Developed and enforced JSON Web Token (JWT) & OAuth2 authentication and role-based access control using **Spring Security**.
- Designed, documented, and validated **RESTful APIs** with **JUnit** testing to guarantee scalability, reliability, and maintainability.

ContactVault | NodeJs, MongoDB, Express - Backend

Mar 2025

- Built a RESTful backend with **Node.js**, **Express.js**, and **MongoDB**, delivering secure Create, Read, Update, Delete (CRUD) operations and optimized NoSQL queries for faster response times.
- Implemented **JWT-based authentication** and role-based access control Role-Based Access Control (**RBAC**) to ensure data security and user privacy.
- Optimized database queries and API response times by applying indexing, aggregation pipelines, and efficient middleware design.

Technical Skills

Languages: Java, C, C++, JavaScript, Python, SQL

Frameworks & Libraries: Spring Boot, Node.js, React.js, Express.js

Databases: MySQL, MongoDB

Tools & Platforms: AWS, Excel, Git, GitHub, PowerBI, Postman, IntelliJ IDEA, VS Code

Certifications

- **Data Visualization using Power BI** - Sharda Informatics — May–June 2025 
- **Prompt Engineering** - Sharda Informatics — Jan–April 2025 
- **AI: Transformative Learning with TechSaksham** - TechSaksham — 2025 
- **Python Pro Bootcamp** - Udemy — 2024 
- **Mastering in C** - Udemy — 2022 